Week 1 | CLO 1-B | File 3 / 7

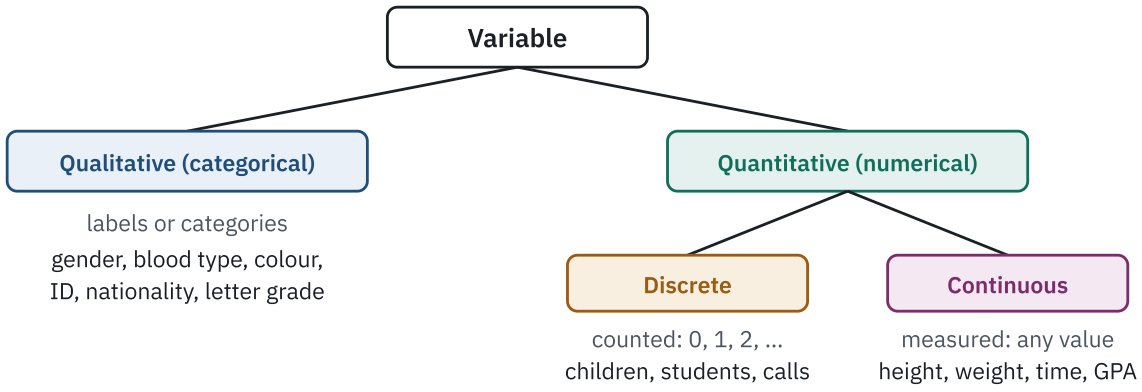
1-2 Variables and Types of Data

المتغيرات وأنواع البيانات — وصفي وكمي، منفصل ومتصل، ومستويات القياس الأربعة

1 · Qualitative vs quantitative

BOTH · CLO1-B

	QUALITATIVE (CATEGORICAL) وصفي / نوعي	QUANTITATIVE (NUMERICAL) كمي / عددي
Definition	Values are labels or distinct categories according to some characteristic or attribute.	Values can be counted or measured.
Examples	colour, type, yes/no, gender, **ID**, major preference (Math, Stat, Physics), phone brand (Samsung, Apple, LG, Huawei), **letter grade** (A–F), nationality, blood type	length, height, age, weight, temperature, volume, area, distance, GPA, exam marks, number of children per family, number of Chemistry students in a class



Ask two questions in order: is the value a category or a number? If a number, was it counted (discrete) or measured (continuous)? A number used as a label, like a student ID, is still qualitative.

اسأل سؤالين بالترتيب: هل القيمة تصنيف أم رقم؟ وإذا كانت رقمًا: هل نحصل عليه بالعدّ (منفصل) أم بالقياس (متصل)؟ الرقم المستخدم كاسم، مثل الرقم الجامعي، يبقى متغيرًا وصفيًا.

2 · Discrete vs continuous

BOTH · CLO1-B

	DISCRETE منفصل	CONTINUOUS متصل
Description	Countable values (0, 1, 2, 3, ...), no fractions or decimals; **obtained by counting**.	Infinitely many values between any two values; fractions and decimals; **obtained by measuring**.

	DISCRETE منفصل	CONTINUOUS متصل
Examples	children in a family; students in a classroom; calls a call centre receives each day; touchdowns scored per season; pens in a box	temperature, weight, GPA, age, height, volume, area, distance, exam marks, air-pollution concentration, time

CONTINUOUS DATA ARE RECORDED ROUNDED, SO A VALUE STANDS FOR AN INTERVAL (TABLE 1-1) AHMED

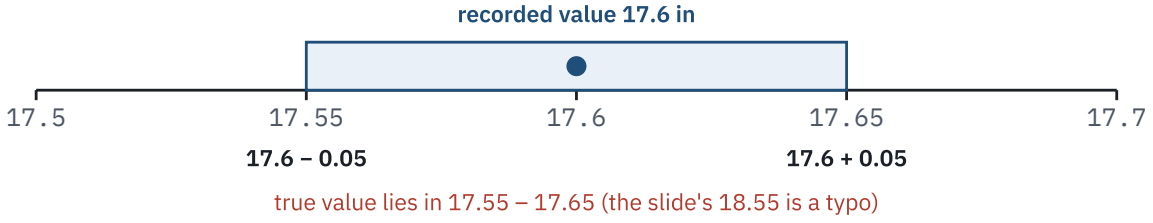
VARIABLE	RECORDED	BOUNDARIES
Length	15 cm	14.5–15.5 cm
Temperature	86 °F	85.5–86.5 °F
Time	0.43 s	0.425–0.435 s
Mass	1.6 g	1.55–1.65 g

Rule: subtract and add **half of one unit of the last recorded digit**. Whole numbers ± 0.5; one decimal ± 0.05; two decimals ± 0.005. The same rule builds class boundaries in Chapter 2.

EXAMPLE 1-3 · CLASS BOUNDARIES AHMED

(a) 17.6 inches (b) 23 °F (c) 154.62 mg/dl

(a) **17.55–17.65 in** (the slide's "18.55" is a typo) · (b) **22.5–23.5 °F** · (c) **154.615–154.625 mg/dl**



A continuous value is rounded when recorded, so 17.6 stands for every true value from 17.55 to 17.65. Go half a unit of the last recorded digit each way: one decimal place means ± 0.05.

القيمة المتصلة تُقرّب عند تسجيلها، فالرقم 17.6 يمثل أي قيمة حقيقية بين 17.55 و 17.65.
نضيف ونطرح نصف وحدة آخر خانة: خانة عشرية واحدة تعني ± 0.05.

EXAMPLE 1-2 · DISCRETE OR CONTINUOUS? AHMED

(a) Hours per week children aged 12–15 reported watching TV. (b) Touchdowns a quarterback scored each year. (c) Money a person earns per week at a fast-food restaurant. (d) Weights of NFL players.

ACTIVITY · VARIABLE TYPE AMANI

Classify, and if quantitative say discrete or continuous: country of birth · hours worked · workers currently employed · full-time or part-time · length of a pen · type of pen · number of pens in a box · colour of the pen's ink.

(a) `continuous`: time is measured. (b) `discrete`: counted. (c) `discrete`: in Bluman, money changes in steps of the smallest unit (cents, or baisa in Oman), so it is countable. (d) `continuous`: measured.

Country of birth `qualitative` · hours worked `quantitative, continuous` · workers employed `quantitative, discrete` · full/part-time `qualitative` · length of pen `continuous` · type of pen `qualitative` · number of pens `discrete` · ink colour `qualitative`.

الشرح بالعربي — نوع المتغير

السؤال الأول: هل القيمة **تصنيف** (ذكر/أنثى، فصيلة دم، جنسية) أم **رقم له معنى حسابي**؟ الأول وصفي (qualitative) والثاني كمي (quantitative). ثم للكمي: هل نحصل عليه **بالعدّ** (عدد الأطفال، عدد الطلاب) فهو منفصل (discrete)، أم **بالقياس** (الطول، الوزن، الزمن، الحرارة، المعدل) فهو متصل (continuous). فخاخ: (1) **الرقم الجامعي** رقم لكنه مجرد اسم، فهو **وصفي**. (2) **التقدير A, B, C** وصفي. (3) **المال** منفصل عند Bluman لأنه يُعدّ بأصغر وحدة (بيسة/سنت). (4) القيمة المتصلة تُسجَّل مقرّبة، فـ 17.6 تعني أي قيمة بين 17.55 و 17.65 .

3 · The four levels of measurement

BOTH · CLO1-B

	Categories	Natural order	Equal differences	True zero (ratios OK)	
Nominal	✓	○	○	○	gender, blood type, ID, nationality
Ordinal	✓	✓	○	○	letter grade, ranking, poor/fair/good
Interval	✓	✓	✓	○	°C, °F, IQ, SAT, calendar year, date
Ratio	✓	✓	✓	✓	height, weight, age, income, time, K

↑ more information

Each level keeps every property of the level below and adds one. To classify a variable, find the highest property it has: zero meaning 'none' makes it ratio; equal steps without a true zero makes it interval.

كل مستوى يحتفظ بخصائص المستوى الأدنى ويضيف خاصية واحدة. لتصنيف متغير ابحث عن أعلى خاصية يملكها: إذا كان الصفر يعني «لا شيء» فهو نسبي (ratio)، وإذا كانت الفروق متساوية بلا صفر حقيقي فهو فتري (interval).

LEVEL	DEFINITION	EXAMPLES (BOTH DECKS + TABLE 1-2)
Nominal (lowest) اسمي	Data are classified into mutually exclusive (nonoverlapping) **categories with no natural order**. Numbers, if used, are just identifiers. No arithmetic; differences are meaningless.	gender, hair colour, phone brand, days of the week, months, names, nationality, student ID, zip code, eye colour, political or religious affiliation, major field, blood type

LEVEL	DEFINITION	EXAMPLES (BOTH DECKS + TABLE 1-2)
Ordinal ترتيبي	Categories that **can be ranked** (natural order), but precise differences between ranks do not exist or are not meaningful (the gap A–B may differ from B–C; two students with grade A can have different scores).	letter grades, superior/average/poor, health status, level of education, Likert scale (strongly agree ... strongly disagree), judging 1st/2nd/3rd, ranking of tennis players, rating poor/good/excellent
Interval فتري	Ranked, with **precise, meaningful differences**, but **no true zero**: zero does not mean absence. Addition and subtraction are valid; ratios are not (140 IQ is not "twice as intelligent" as 70).	temperature in °C or °F, IQ, SAT score, dates (production/expiry, birth date), years (graduation 2018), exact clock time (leaving home at 8:35 AM)
Ratio (highest) نسبي	All interval properties **plus a true zero** (0 means none). All arithmetic is valid, ratios included: 20 kg is twice 10 kg.	height, weight, area, age, salary or income, number of phone calls, exam marks, GPA, time duration (45 minutes), temperature in Kelvin

EXAMPLE 1-4 AHMED

(a) ages of authors of the top 25 fiction books (b) colours of baseball hats sold (c) highest temperature each day of a month (d) ratings of bands in a parade

(a) `ratio` (b) `nominal` (c) `interval` (d) `ordinal`

ACTIVITY · LEVEL OF MEASUREMENT AMANI

(1) blood types of 500 patients (2) Muscat temperatures 25, 30, 35 °C (3) heights 150, 165, 180 cm (4) nationalities of tourists (5) race places 1st, 2nd, 3rd (6) fitness-tracker steps per day (7) years of historical events 1990, 2005, 2023

(1) `nominal` (2) `interval`: 0 °C is not "no temperature" (3) `ratio` (4) `nominal` (5) `ordinal` (6) `ratio`: 0 steps means none (7) `interval`: year 0 is not "no time"

Active-learning task (first 5 minutes of the next lecture): use an AI tool for 3 minutes to find an example of each level, then in pairs one student gives an example and the other names the level.

الشرح بالعربي — مستويات القياس

رتّبها من الأدنى للأعلى، وكل مستوى يأخذ خصائص الذي قبله ويزيد واحدة: (1) **اسمي**: تصنيفات بلا ترتيب (الجنس، فصيلة الدم، الرقم الجامعي). (2) **ترتيبي**: تصنيفات لها ترتيب لكن المسافات بينها غير محددة (التقديرات، المراكز الأولى والثانية، ممتاز/جيد/ضعيف). (3) **فتري**: فروق متساوية لها معنى لكن **لا يوجد صفر حقيقي** (الحرارة بالمئوية، السنة، التاريخ، الذكاء IQ). (4) **نسبي**: يوجد **صفر حقيقي** يعني «لا شيء»، فتصح النسب (الطول، الوزن، العمر، الراتب، الزمن المستغرق، عدد الخطوات). الاختبار الحاسم بين الفتري والنسبي: هل الصفر يعني **انعدام** الشيء؟ 0 كجم = لا وزن ← نسبي. 0 درجة مئوية ليست «لا حرارة» ← فتري.

BORDERLINE CASES EXAMINERS USE

Year / date / clock time: interval, but **time duration** (45 minutes) is ratio. **Temperature**: °C and °F are interval, Kelvin is ratio. **Numeric codes** (ID, zip code, phone number): nominal and qualitative. **Exam marks**: ratio in Dr. Ahmed's list, but **letter grades** are ordinal. **IQ and SAT**: interval.

Chapter 1 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief · 01 · Lesson 1-1: Descriptive and Inferential Statistics · **02 · Lesson 1-2: Variables and Types of Data** · 03 · Lesson 1-3: Data Collection and Sampling · 04 · Revision pack · 05 · Exam-style questions · 06 · Full worked answers